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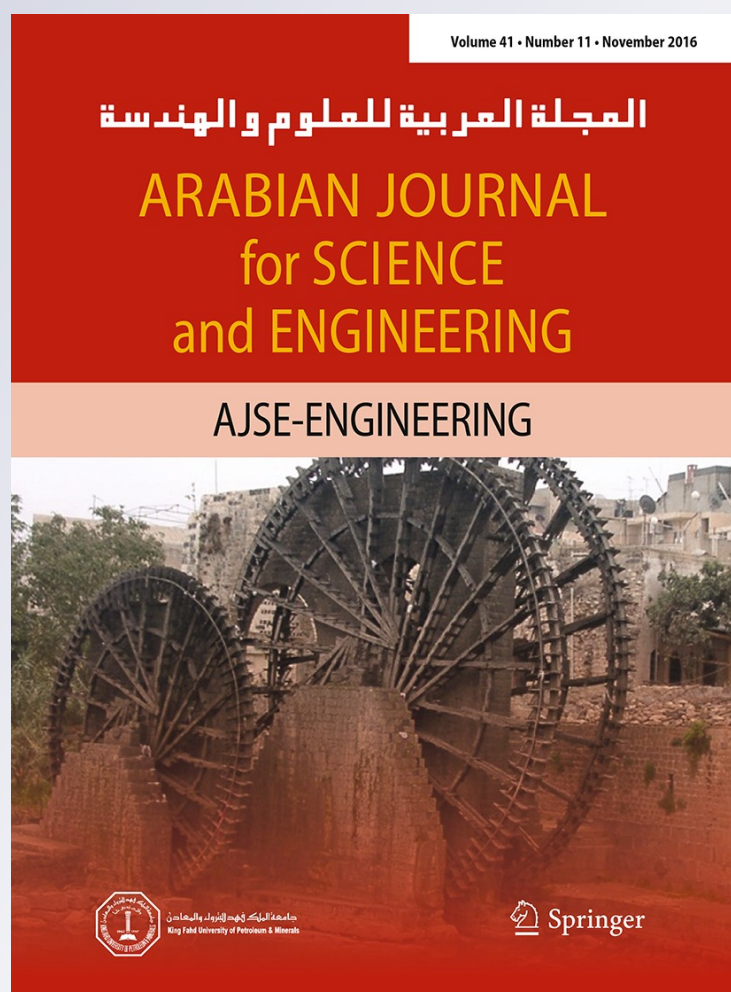
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Investigation on Mechanical and Tribological Behaviors of PA6 and Graphite-Reinforced PA6 Polymer Composites

S. Sathees Kumar¹ · G. Kanagaraj²

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Abstract In this paper, the effects of adding graphite on the mechanical and tribological properties of polyamide 6 (PA6)-based composites were investigated. PA6 was reinforced with graphite by varying the weight proportions. The test specimens for mechanical and tribological experiments were molded in an injection molding machine. The mechanical properties of PA6 and graphite-reinforced PA6 polymer composites were investigated in terms of tensile strength, impact and hardness tests. The PA6 composite containing 20 wt% graphite revealed the better mechanical behaviors. The wear test was observed by using a pin-on-disk machine under dry sliding conditions at 5, 10, 20 and 30 N applied load values and sliding speed of 1000, 1500 and 2000 rpm. Scanning electron microscopy was utilized to examine the worn surfaces microstructure and wear mechanisms before and after the tribological test. The results showed that the reinforced PA6 polymer composites with 20 wt% graphite enhanced the best tribological behavior. The addition of graphite enhances the life condition of normal PA6 to a greater extent. Also, this composite exhibits stronger interfacial bonding characteristics with improved wear resistance.

Keywords Polyamide · Reinforced polymer · Graphite · Wear · Friction

1 Introduction

Polyamide 6 (PA6) and its composites have been increasingly used in various ranges of industrial applications such as aerospace, automotive, electronics and chemical industries. This is because these materials have high strength/weight ratio in comparison with classic material, self-lubricant property, good damping property, corrosion resistance, UV and gamma radiation resistance and relatively simple and economic fabrication process [1]. However, limitations in mechanical properties level, low heat deflection temperature, high water absorption and dimensional instability of pure PA6 have prevented its wide range of applications. The fact that polymer materials have much lower mechanical strengths at elevated temperatures, lower thermal conductivities and higher coefficients of thermal expansion makes them highly sensitive to temperature changes [2]. To overcome this difficulty, many researchers paid attention and improved the property of PA6 significantly by using reinforcement methods. Reinforcements are usually done by adding solid lubricants, fillers and additive materials to improve the properties of the base metals like strength, stiffness and conductivity.

The most commonly used are molybdenum disulfide, graphite, fiber and glass beads and various ceramic particles. Graphite (Gr) powders are widely used as fillers to improve the electrical, thermal and mechanical properties of polymers; their poor affinity with polymers has limited their industrial applications in many fields [3]. Graphite is naturally abundant and low cost and has been used as electronically conducting filler in preparing conducting polymer composites [4]. Suresha et al. [5] and Kalaitzidou et al. [6] reported that graphite can not only reduce the friction coefficient and wear rate, but also enhance the thermal conductivity and the mechanical properties of the polymer matrix. In this study, the influence of graphite on mechanical and tribolog-

✉ S. Sathees Kumar
shrutishyami@gmail.com

G. Kanagaraj
gkmech@tce.edu

¹ Department of Mechanical Engineering, Latha Mathavan Engineering College, Madurai, Tamil Nadu, 625301 India

² Department of Mechanical Engineering, Thiagarajar College of Engineering, Madurai, Tamil Nadu, 625015 India



ical properties of pure polyamide-6 (PA6), 5, 10, 20, 30 and 40 wt% of graphite-reinforced PA6 polymer composites was investigated.

2 Literature Review

In the past few years, various kinds of polymer composites with remarkable enhancement in mechanical and thermal properties have been reported [7–9]. The mechanical and tribological behavior of polyamide, high-density polyethylene, and their composites have been studied and reported in the literature [10–12]. Stokes et al. [13] investigated the mechanical properties of fiber-reinforced PA6 composites fabricated using an injection molded method. The friction and wear characteristics of monolithic polyamide 66 (PA66) and PA66 composite reinforced with short glass fibers were studied in [14]. They also reported that PA66 containing 40 vol% glass fibers improved the wear resistance.

Palabiyik et al. [15] observed the mechanical and tribological properties of PA6 and high-density polyethylene, polyblends with and without the compatibilizing agent maleic anhydride propylene, and they found that the best coefficient of friction and high wear resistance was identified in the composites consisting 40 % of PA6 and 60 % of HDPE and 80 % of PA6 and 20 % of HDPE, respectively, by weight. By further investigating the PA6 and HDPE polyblend compositions, the maximum reduction in wear and the coefficient of friction were obtained by filling the polyblends with 10 % of polytetrafluoroethylene (PTFE) by weight [16]. Bin et al. [17] reported the friction and wear properties of polyamide 66 (PA66), polyphenylene sulfide (PPS) and PTFE by conducting wear test using a pin-on-disk tribometer under dry sliding and oil-lubricated conditions. They reported that the friction and anti-wear properties of PTFE and PPS were improved by oil lubrication while that of PA66 were decreased by oil lubrication.

Cong et al. [18] prepared PA46/ HDPE polyblends with different volume ratios of 8/2, 6/4, 4/6 and 2/8. They used scanning electron microscopy (SEM), differential scanning calorimetry (DSC) and X-ray diffraction (XRD) for investigating the phase structure and crystalline characteristics of polyblends, and the results showed that the friction coefficient of all PA46/HDPE blend was as low as 0.21–0.22, which were near the friction coefficient of pure HDPE, and much lower than that of pure PA46. Li et al. [19] examined reinforced PA6 with polyurethane (PA6-PU) block copolymers in different short glass proportions and found that the addition of glass fiber content was 5 %, the coefficient of friction decreased by 30 % and the wear rate decreased by 68 %. The experimental study conducted by [20] on mechanical and tribological properties of polytetrafluoroethylene PTFE/PA6 blends with different compositions (0, 10, 20, 30

and 40 vol%) and reported that the optimum value of wear reduction for the composite content was 30 vol% of PA6.

The effects of incorporating montmorillonite (MMT) with PA6 composite were studied in [21], and the composite was prepared by twin screw extruder mixer technique. The incorporation of MMTs (except 5 % by weight) into PA6 effectively enhances the wear resistance, and the composites containing 3 wt% of MMTs in the PA6 matrix exhibit the best performance in tensile strength and flexural strength. Pan et al. [22] analyzed the mechanical and tribological properties of monomer casting (MC) nylon/graphene oxide nanocomposites, and they found that the nanocomposites have lower wear rates in comparison with MC nylon. The possibility of improving the tribological property and thermal conductivity of PA6 by the addition of copper and graphite was investigated in [23].

Li et al. [24] have demonstrated the tribological properties of the glass fiber-reinforced PA6 (GF/PA6, 15/85 by weight) for high-performance friction materials, using single or combinative solid lubricants of polytetrafluoroethylene (PTFE), ultra-high molecular weight polyethylene (UHMWPE) and the combination of both of them. Li et al. [25] examined the thermal conductivity, mechanical properties and heat deflection temperature of the magnesium hydroxide ($\text{Mg}(\text{OH})_2$), alumina (Al_2O_3) and flake graphite-filled PA6 composites. The preparation process of PA6 composite with graphite powder was illustrated in [25]. The graphite powder was first oxidized with hydrogen peroxide (GPs-OH) and then modified with hexamethylene diisocyanate (GPs-NH₂). In this work, X-ray photoelectron spectroscopy (XPS), elemental analysis, thermogravimetric analysis (TGA), X-ray diffraction (XRD) tests, tensile and dynamic mechanical analysis (DMA) were carried out and found that the mechanical properties of GPs-NH₂ were improved because of the good affinity between the modified GPs and PA. Demir [26] investigated the sliding wear performance of 20% mica-filled polyamide (PA6 + 20 % mica) and 20 % short glass fiber-reinforced polysulfone (PSU + 20GFR) polymer composites used in electrical applications.

Jia et al. [27] studied the effects of adding nanoscale lamellar structure expanded graphite (nano-EG) on the friction and wear properties of hot-molded polyimide-based composites. They also found that the best tribological properties occurred when the nano-EG content is 15 wt%. Ozerol et al. [28] examined the synthesis and characterization of composite materials, based on graphite (GH) and boric acid-doped polyaniline (PANI). PANI + GH composites were prepared via a simple in situ polymerization of aniline in graphite dispersion. Gong et al. [29] investigated the covalent functionalization of graphene oxide (GO) with polyvinyl alcohol (PVA) via ester linkages (GO-es-PVA) as well as the characterization of modified graphene-based Nylon-6 composite prepared. The efficient polymer chain grafting, a homoge-



neous dispersion of GO sheets in PA6 matrix and a dramatic improvement of interface adhesion between nanosheets and matrix were observed in PA6/GO-es-PVA composites by SEM and TEM. Faghihi et al. [30] studied the effects of molecular weight and structure of PA6 morphology and properties of PA6/MWCNT composites. Fasahat et al. [31] studied the wear properties of PA6 nanocomposite fabrics, and melt-spun yarns were filled with different concentrations of Ag/SiO₂, TiO₂, ZnO and their mixtures. Zhang et al. [32] investigated the tribological properties of graphite (Gr) or carbon nanotubes (CNTs) or both of them were incorporated into carbon fabric-reinforced phenolic (CFRP) composites. The tribological tests were conducted under different sliding speed, load and oil lubrication conditions.

The above literature reveals that there are only very few articles that have investigated the effect of graphite content in PA6 with graphite-reinforced PA6 composites. Nevertheless, the tensile properties, friction and wear characteristics of graphite-reinforced PA6 composites have not been suggested. In this study, the mechanical and the tribological behavior of PA6, graphite-reinforced PA6 composites were investigated by conducting tensile, impact, hardness and wear tests. The morphological images of before and after wear test specimens were examined in SEM for further analysis.

3 Experimental Details

3.1 Materials and Instrumentation

Polyamide 6 (PA6) purchased as pellets of size 3 mm, whose tensile strength was 80 MPa, flexural strength 105 MPa, density 1.14 g/cm³ and melting temperature 220 °C, respectively. Graphite fine powder of 98 % size and 60 meshes were used as a reinforced materials for PA6 polymer composites. The test specimens were prepared using an injection molding machine with maximum chamber capacity of 150 g. The muffle furnace was supplied by Prabha machine tools, Coimbatore, Tamil Nadu, India. The tensile test was conducted on a UTM DTRX 10 kN Electronic testing machine manufactured by Deepak Polyplast Pvt. Ltd. Ahmedabad, India. The impact test was conducted on a Tinius Olsen tester. The hardness test was conducted on a Shore D hardness test machine (Model-SRT-102) manufactured by S. C. Dey & Co. Kolkata, India.

3.2 Preparation Process of PA6 and Graphite-Reinforced PA6 Composites

In order to evaluate the mechanical and tribological properties of PA6 blended with graphite, the blends were produced in different proportions of PA6 and graphite. In the test specimen preparation process, the graphite was added

to PA6 with 5, 10, 20, 30 and 40 wt% ratios. The graphite particles with high melting point do not blend with PA6 in right proportions. Therefore, graphite was preheated below melting temperature. The preheating was carried out in a muffle furnace maintained at the temperature of 2320 °C. After reaching the temperature of 2320 °C the preheated graphite was added with PA6 pellets for through mixing. The specimens of PA6 and graphite-reinforced PA6 composites were molded by using an injection molding machine at temperatures from 230 to 250 °C. Figure 1 shows the preparation process of test specimen using an injection molding machine. The Resistance temperature detector (RTD) sensor is inserted in the circumference of the injection molding machine for temperature measurement. The sensing temperatures of PA6 material were observed using digital indicator. The molding die was preheated up to 75 °C for the free flow of melted composites. The temperature of mixtures plays an important role in the composite materials, because heating improves the blending nature of polymers with other materials.

3.3 Specifications of PA6 Polymer Composites

The test specimens for mechanical and tribological experiments were molded in the following composition (given in % weight) using an injection molding machine.

- 100 wt% PA6
- 95 wt% PA6 + 5 wt% Gr
- 90 wt% PA6 + 10 wt% Gr
- 80 wt% PA6 + 20 wt% Gr
- 70 wt% PA6 + 30 wt% Gr
- 60 wt% PA6 + 40 wt% Gr

Figure 2 shows the image of PA6 and graphite-reinforced PA6 specimen obtained from injection molding machine. The natural properties of PA6 and graphite which are used to form the reinforced polymer are given in Table 1.

3.4 Testing Conditions and Measurement

The PA6 and graphite-reinforced PA6 composite test specimen were cut into required shape and machined according to test standards. The tensile test was conducted on a UTM DTRX computer-controlled Universal Testing Machine at room temperature. The crosshead speed of the machine was set at 5 mm/min. The specimen size for the tensile test was 115 mm long, 20 mm wide and thickness 3 mm according to ASTM D638 standard. The measurements taken at five constant loads for five specimens were recorded. Tensile strength and elongation values were obtained from the load–displacement curve.



Fig. 1 Preparation process of PA6 and graphite-reinforced PA6 test specimen

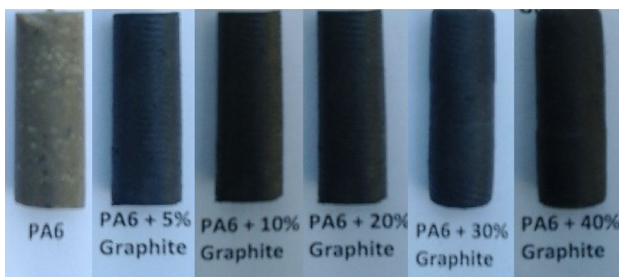
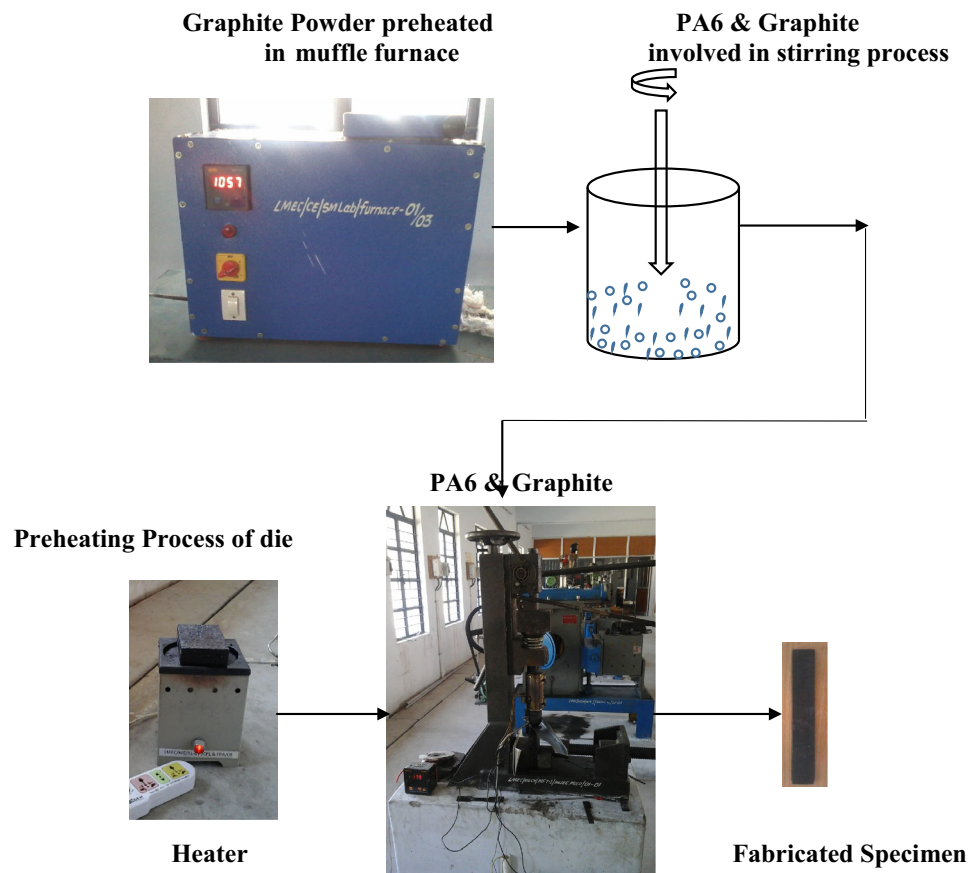


Fig. 2 Image of PA6 and graphite-reinforced PA6 test specimen

Table 1 Mechanical properties of PA6 and graphite

Materials	Tensile strength (MPa)	Tensile modulus (MPa)	Flexural strength (MPa)	Hardness
PA6	80	3000	105	82 (Shore - D)
Graphite	76	27600	345	VHN ₁₀ 7-11 (kg/mm ²) 107.87315 (MPa)

The impact test was conducted on a Tinius Olsen tester according to ASTM D256 at room temperature. The specimen size for the impact (Izod) test was $64 \times 12.7 \times 3.2$

mm. The hardness test was conducted on a Shore D hardness test machine. The specimen size for hardness test was 35 mm length, 15 mm width and thickness 3 mm according to ASTM D 2240 standard.

The tribological test was carried out on a TR-20LE-PHM-400, DUCOM tribometer. Before applying the test, the test specimens were cleaned with methanol to remove the possible impurities on the surface. The test specimen was shaped for ASTM G99 standard and machined as 12 mm diameter and 30 mm length. Tests were carried out at sliding speeds of 1000, 1500 and 2000 rpm and applied load values of 5, 10, 20, and 30 N. Throughout the test, the friction and wear data were collected by the data acquisition system stored and exhibited on a computer screen. Furthermore, to ensure the reliability of the results, each test was repeated thrice. At least five samples were tested to obtain average values and were recorded for further analysis. In order to obtain an accurate value of weight loss, the samples were placed in an atmospheric air for 24 h to remove the moisture content. The weights of all test specimens were measured using a precision electronic balance. Morphologies of fractured composite surfaces were examined by scanning electron microscopy. The constraints and test conditions used in the pin-on-disk testing machine are given in Table 2.

Table 2 Constraints and testing conditions

Test specimen	Constraints	Testing conditions
PA6	Load	5, 10, 20 and 30 N
PA6 + 5 wt% Gr	Speed	1000–2000 rpm
PA6 + 10 wt% Gr	Pin materials	PA6 with graphite
PA6 + 20 wt% Gr	Surface condition	Dry friction
PA6 + 30 wt% Gr	Track diameter	100 mm
PA6 + 40 wt% Gr	Disk material	EN31 Steel
	Sliding distance	1000 m

3.5 Heat Calculation

If the load is increased at a higher velocity, it is known that the surfaces of polymers melt at a certain load because of friction heating. Normally, the heat generation occurs in sliding between the materials and temperature increases between the two friction surfaces. The amount of heat generated can be determined by $H = \mu PV$ (watts) where μ is the coefficient of friction, P is the applied load and V is the sliding velocity. The calculations for heat generated during wear tests for the loads 5, 10, 20 and 30 and for various speeds 1000, 1500 and 2000 rpm are obtained and are given in Table 3. Friction heat is hard to dissipate and promotes adhesive and fatigue wear [31]. The specimen temperature increases with the increase in the applied load and the duration of the test [32]. As compared to PA6, the composite material shows lower interfacial temperature due to the high thermal conductivity of the filler materials.

3.6 SEM Analysis of Before and After Worn Surface

In order to compare and observe the morphological structures, the SEM images were taken before and after wear test specimens. Figure 3 shows the SEM images of PA6 and graphite-reinforced PA6 composites test specimen's surfaces taken in before and after wear test conditions. Figure 3a, c, e, g, and i shows the SEM images taken before the wear test

for PA6, PA6 + 5 wt% Gr, PA6 + 10 wt% Gr, PA6 + 20 wt% Gr, PA6 + 30 wt% Gr and PA6 + 40 wt% Gr, respectively. Figure 3b, d, f, h, and j shows the SEM images taken after the wear test for PA6, PA6 + 5 wt% Gr, PA6 + 10 wt% Gr, PA6 + 20 wt% Gr, PA6 + 30 wt% Gr and PA6 + 40 wt% Gr, respectively.

Figure 3a and b reveals the morphological image of pure PA6 test specimen before and after worn surfaces. In Fig. 3c, the darker particles represent the graphite and the lighter ones are PA6. The bonding structure of PA6 and graphite particles is formed layer by layer due to small amount of graphite particles reinforced with PA6. The bonding strength of graphite is clearly visible in Fig. 3d which shows the migration action of PA6 and graphite. Figure 3e reveals that the graphite particles gradually penetrate into the PA6 due to added graphite content by 10 wt%. The amount of graphite particle is increased beyond 10 % weight, and there is some accumulation and cluster of the particles (Fig. 3f). The cluster formation occurs on the relative frictional surface. Also, it shows that fragmentation is found out on the surface of PA6 with 10 wt% graphite. Here, the filler material dislocated due to maximum applied load and created higher friction on PA6. Figure 3g shows the uniform dispersion of graphite particles with PA6. The SEM image (Fig. 3g, h) of 20 wt% graphite-reinforced PA6 displays the uniform distribution of graphite particles in PA6. It is clearly observed a good interfacial bonding between graphite and polymer material matrix. The worn surface of PA6 + 20 wt% graphite composite was relatively smooth, a few fine wear debris appeared on the worn surface (Fig. 3h). The graphite has good bonding to the polymer matrix and supports the load from the counter face effectively [23]. However, the surface morphology of graphite-reinforced PA6 composites is formed to display a very smooth pattern of PA6 with 20 wt% graphite (Fig. 3h). It reveals that the graphite particles strongly protect the top of the surface of the composite material under dry sliding test conditions.

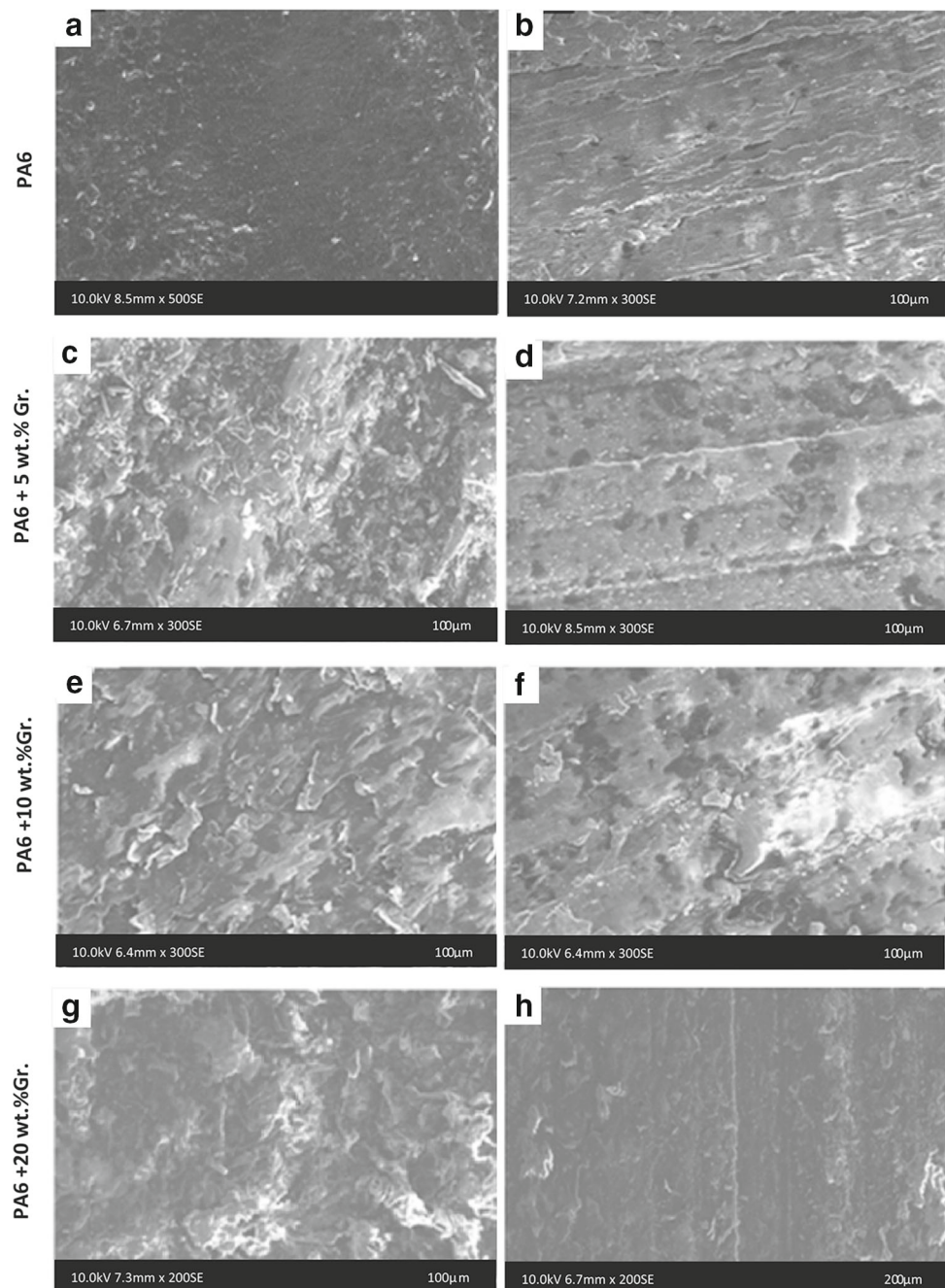
Figure 3i and j shows the SEM image of 30 wt% graphite-reinforced with PA6 composites. By further adding the

Table 3 The amount of heat generation (watts) for PA6 and graphite-reinforced PA6 composites test specimen during wear test conditions

Test specimen	Load 5N			Load 10N			Load 20N			Load 30N		
	Sliding velocity (m/s)			Sliding velocity (m/s)			Sliding velocity (m/s)			Sliding velocity (m/s)		
	5.23	7.85	10	5.23	7.85	10	5.23	7.85	10	5.23	7.85	10
PA6	7.8	12.9	18.5	19.7	27.4	37	39.5	58	78	86.4	91.8	113
PA6 + 5 wt% Gr	7.3	11.7	16	18.6	25.9	35	36.4	54.9	76	83.3	89.7	107
PA6 + 10 wt% Gr	7.5	12.5	17.5	19.2	26.6	36	37.7	55.3	77.3	84.9	90.7	109
PA6 + 20 wt% Gr	6.7	11	16	17.1	25.9	32.8	33.3	51.8	68	80.2	84	99
PA6 + 30 wt% Gr	6.9	11.3	15.4	14.6	24	33.4	34.9	53.6	71.5	82.7	85.3	102
PA6 + 40 wt% Gr	7.2	11.8	15.9	17.6	25.3	32.4	35.7	55.4	73.5	83.9	86.8	105



Fig. 3 SEM images of PA6 and graphite-reinforced PA6 composite test specimen surface under before and after wear test conditions (a, c, e, g, i and k before wear test and b, d, f, h, j and l after wear test)



graphite content, the worn surface of PA6/graphite is slightly smooth (Fig. 3i), and there is some voids which are found on the surface of PA6. The poor interfacial bonding is observed in Fig. 3j due to the uneven distribution of graphite and improper mixing results PA6 detached from the graphite particles. Figure 3k and l shows the SEM image of 40 wt% graphite-reinforced PA6 composite test specimen's surface under before and after wear test conditions. With the addition of graphite content before worn surface is not much smooth (Fig. 3k), the after worn surface of PA6/graphite indicates the crack propagation of the matrix, deterioration

of the polymer–matrix adhesion due to repetitive mechanical stress and some particles pull out from the matrix also visible (Fig. 3l). It indicates the deterioration of the polymer–matrix adhesion due to repetitive mechanical stress and debonding of polymers from the matrix. Figure 3c–l shows the graphite particles integrated on the PA6 composites, and consequently the worn surface is protected from the wear and friction. The uniform distribution of the graphite particles in the microstructure of the PA6 material is the major responsible factor for the improvement in tribological properties of PA6.



Fig. 3 continued

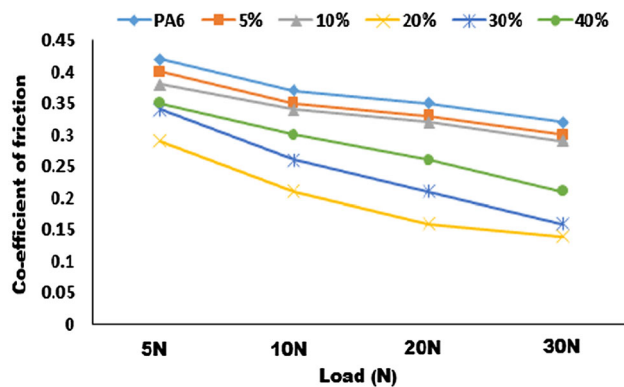
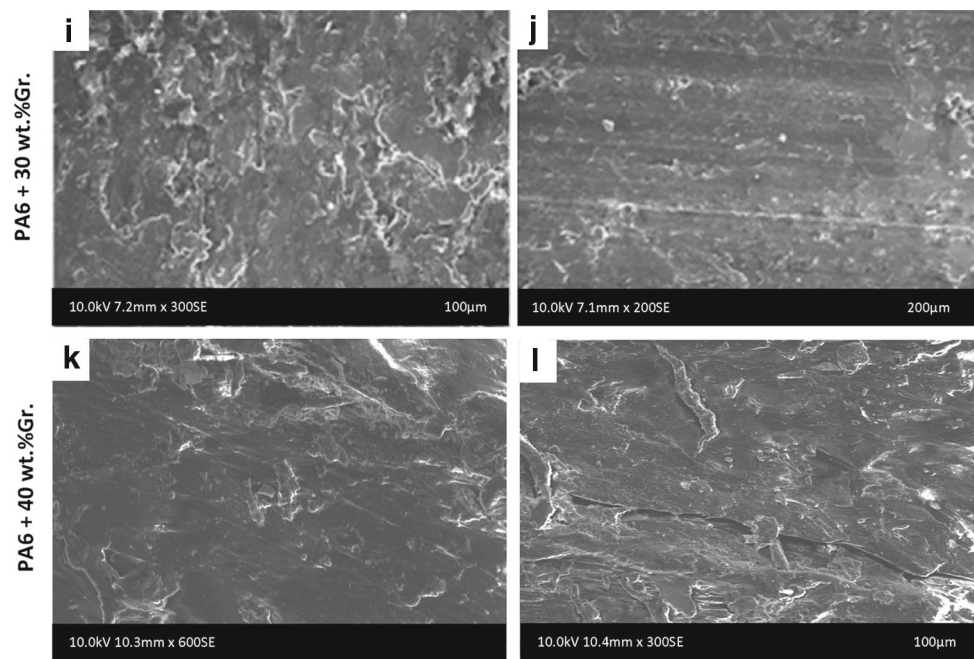


Fig. 4 Variations of the friction coefficient for PA6 and graphite-reinforced PA6 composites

4 Results and Discussion

4.1 Performance of Friction Coefficient

Figure 4 shows the variation of frictional coefficient of PA6 and graphite-reinforced PA6 composites for different loads. The decrease in friction co-efficient with an increase in load is attributed because of maximum sliding velocities at the different weight fraction of reinforcement. The graphite-reinforced PA6 composite materials have lower frictional co-efficient as compared to pure PA6. However, it was observed during the experiment that the composite material shows a slight change in frictional coefficient with the variation of load and sliding velocity. The average coefficient of frictions is in the range of 0.4 and 0.3 for PA6 with 5 wt% graphite composite, 0.38 and 0.29 for PA6 with 10 wt% graphite com-

posite and the average coefficient of frictions are in the range of 0.29 and 0.14 for PA6 with 20 wt% graphite composite, the average co-efficient frictions are in the range of 0.34 and 0.16 for PA6 with 30 wt% graphite composite. As a final point, the average co-efficient frictions are in the range of 0.35 and 0.21 for PA6 with 40 wt% graphite composite. However, the average co-efficient of friction is 0.42 and 0.32 for pure PA6 material. There is a significant reduction in the friction coefficient at graphite weight content as low as PA6 with 20 wt% graphite. Taking all these results into consideration, it was understood that the presence of graphite appeared to make effects on the friction co-efficient for the PA6. This is consistent with the results reported in [10]. In addition, the PA6 with a content of graphite 20 wt% showed the lowest co-efficient of friction among the composites tested in this work.

4.2 Performance of Wear Rate

During initial tests, the surfaces of both the specimens and the steel counterparts were rough and thus strong interconnections between the surfaces resulting in a high friction coefficient. As the wear process continued, the rough profiles of the steel counterparts and the specimens were smoothened. When the applied load is small, the stress in the contact region between the specimen and the counterpart is small, the frictional heat also less. With the increase in sliding velocity, the friction heat is high and additional wear debris adheres on the specimen. The specific wear rate (mm^3/Nm) was calculated using the following relation: $\text{Wear rate} = (\Delta m / \rho F d)$, Where Δm is the mass loss in grams, ρ is the density of the test material in g/cm^3 , F is the load in Newton and d is the



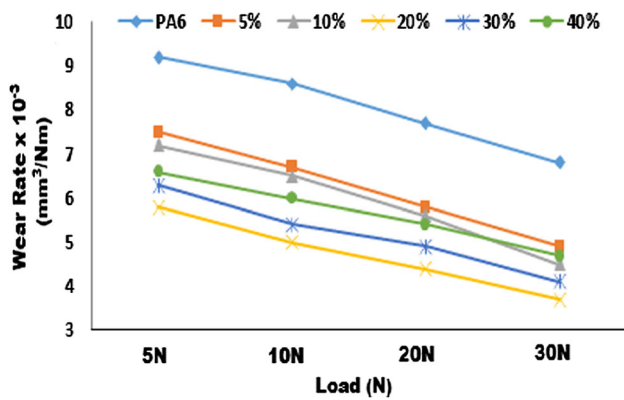


Fig. 5 Variations of the wear rate for PA6 and graphite-reinforced PA6 composites

sliding distance in meters. The readings were noted at a regular time interval of 5 min and were plotted. The material the specimen surface was measured using a precision electronic balance with an accuracy of ± 0.01 mg.

Figure 5 shows the wear characteristics of PA6, graphite-reinforced PA6 composites under 5, 10, 20 and 30 N load. Applied load is one of the most important factors affecting the wear rate of composites. With the increase in applied load, the adhesive wear became dominant. The wear rate of the composite material is not significant changes in pure PA6 material at all loads. This is due to the self-lubricating property of graphite present in the PA6 matrix. In wear test, while the specimen contacts with the disk heat and wear have produced on the contact surface of the specimen. The sliding contact of two materials results in heat generation at asperities and hence increases the temperature [22].

The average wear rate in the range of 7.5×10^{-3} and 4.9×10^{-3} mm³/Nm for PA6 with 5 wt% graphite composite, 7.2×10^{-3} and 4.5×10^{-3} mm³/Nm for PA6 with 10 wt% graphite composite and the average wear rate are in the range of 5.8×10^{-3} and 3.7×10^{-3} mm³/Nm for PA6 with 20 wt% graphite composite and the average wear rate in the range of 6.3×10^{-3} and 4.1×10^{-3} mm³/Nm for PA6 with 30 wt% graphite composite. As a final point, the average wear rate in the range of 6.6×10^{-3} and 4.7×10^{-3} mm³/Nm for PA6 with 30 wt% graphite composite. However, the average wear rate is 9.2×10^{-3} and 6.8×10^{-3} mm³/Nm for pure PA6 material. Hence, PA6 with 30 wt% graphite composite formed high wear resistance and low co-efficient of friction on the surface compared with PA6 and other proportions. The amount of wear decreased with increase in the content of graphite particles. When the hard particles are strongly bonded with a base material, they protect the surface against severe destructive action by the counter face, because of the stronger interfacial bonding characteristics, which plays a vital role in transferring loads from the matrix to the graphite particles. Figure 5 clearly shows that the incorporation of graphite in

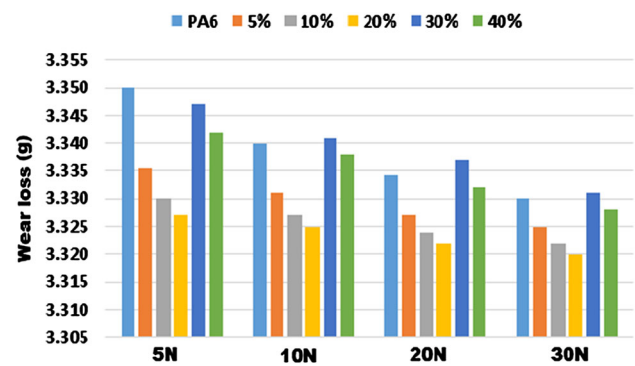


Fig. 6 Variations of the wear loss for PA6 and graphite-reinforced PA6 composites under different loading conditions

PA6 substantially improves the wear resistance characteristics and PA6+20 wt% Gr. Composite exhibits lower wear rate compared with other composites. Further additions of graphite content contribute to a negative effect on the wear and friction coefficient properties of this composite.

4.3 Comparison of Wear Loss for PA6 Composites

Figure 6 shows the wear loss characteristics of PA6 and graphite-reinforced PA6 composites under the applied load of 5, 10, 20 and 30 N. The average wear loss in the range of 3.336–3.325 g for PA6 + 5 wt% graphite composite, 3.330–3.322 g for PA6 + 10 wt% graphite composite, 3.327–3.320 g for PA6 + 20 wt% graphite composite, 3.347–3.331 g for PA6 + 30 wt% graphite composite and 3.342–3.328 g for PA6 + 40 wt% graphite composite, respectively.

The highest wear loss occurred in pure PA6, whereas PA6 + 20 wt% Gr composite material exhibits lesser wear loss. The amount of wear decreased with increase in the content of graphite particles [10]. When the particles are strongly bonded with the base material, they protect the surface against severe destructive action by the counter face and which plays a vital role in transferring loads from the matrix to graphite particles [15,16]. Figure 6 clearly shows that PA6 + 20 wt% Gr has a low wear loss (76.66 %) compared with other proportions. This suggests that the incorporation of graphite in PA6 significantly reduced wear rate, which match the results reported in [25]. The sliding contact of two materials results in heat generation at asperities and hence increases the temperature. The heat dissipation rate is gradually increased due to the reinforcement of graphite with PA6 [14]. During wear test, the amount of heat generated between the contact surfaces. The accumulated heat of the material should be dissipated properly. Otherwise, it will give the negative effect. Graphite particles have excellent antifriction characteristic which to act dissipate the heat from the contact surface gradually.

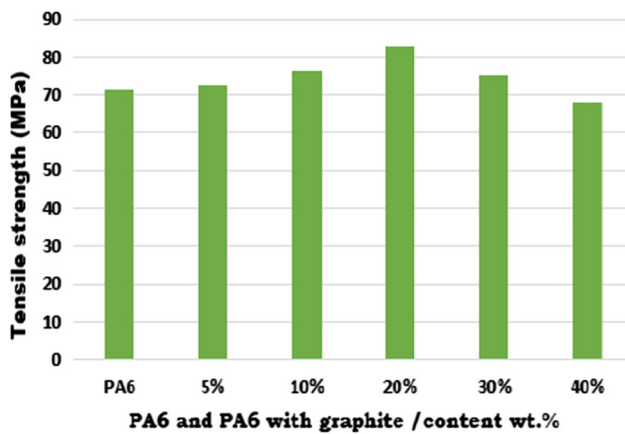


Fig. 7 Effects of graphite content on the tensile strength of graphite-reinforced PA6 composites

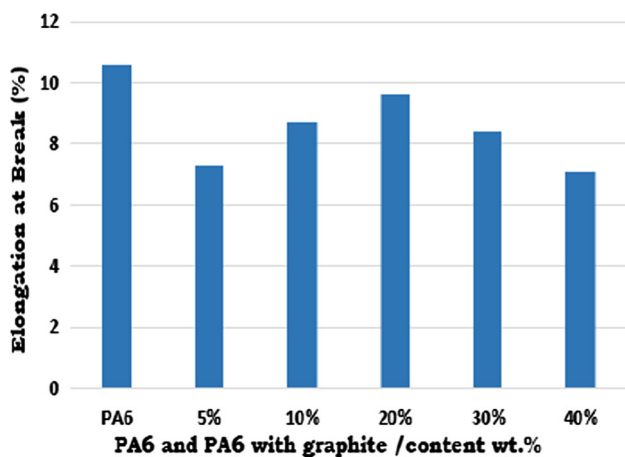


Fig. 8 Variation of elongation at break of graphite-reinforced PA6 composites

4.4 Comparison of Tensile Properties for PA6 Composites

Figures 7 and 8 show the tensile and elongation properties of PA6 and graphite-reinforced PA6 composites. It is seen that tensile strength was improved evidently with the addition of graphite in the PA6 composites. The tensile strength of the PA6 composites increases first and then decreases with the increasing content of Gr; the maximum tensile strength is 83 MPa when the Gr content is 20 wt%. This behavior is

attributed to the high strength of Gr and the poor affinity between Gr and PA6 that led to brittle failure of composite, as seen in SEM images when the Gr content increased. Low tensile strength was observed in PA6 + 40 wt% Gr. According to tensile properties of PA6, graphite-reinforced PA6 composites (Table 4), the reinforcing effect of graphite is significant: PA6 + 20 wt% Gr composites achieve the maximum increase in the tensile strength (16.47 %). The enhanced tensile strength can certainly be attributed to the strong interfacial adhesion and good dispersion of graphite in the PA6 matrix, resulting in effective load transfer from the matrix to graphite.

Figure 7 indicates that the tensile strength of pure PA6 is 71.27 MPa where as PA6 reinforced 20 wt% graphite is 83 MPa. Similar kind of results reported and observed by the authors [23]. Uniform dispersion and the better interfacial adhesion between the dispersed phase and the matrix are the important factors that dominate the tensile properties of the composites. The mechanical properties of PA6/graphite were improved because of the good affinity between the graphite and PA6 [24]. The tensile strength of PA6 with 20 wt% graphite composite was improved by 16.47 % compared with PA6. It shows an excellent ductile property. From the results, the improved tensile properties and elongation strength were noticed for 20 wt% graphite-reinforced PA6 composites compared with other composite proportions. Figure 8 indicates the elongation behavior of PA6 and graphite-reinforced PA6 composites. The elongation at break of PA6 composites decreases first and then increases with the increasing content of Gr; the maximum elongation at break is 9.6 % when the Gr content is 20 wt%.

4.5 Comparison of Impact Properties for PA6 Composites

Figure 9 shows the impact strength of the PA6 and graphite-reinforced PA6 composites. It is observed that the impact strength increased obviously with that addition of graphite content to PA6. The impact strength of the graphite-reinforced composites increases first and then decreases with the increasing content of Gr; the maximum impact strength is 23 kJ/m² when the GPs content is 20 wt%. Further addi-

Table 4 Physical and mechanical properties of PA6 and graphite-reinforced PA6 polymer composites

Properties	Materials					
	PA6	PA6 + 5 wt% Gr	PA6 + 10 wt% Gr	PA6 + 20 wt% Gr	PA6 + 30 wt% Gr	PA6 + 40 wt% Gr
Tensile strength (MPa)	71.217	72.6	76.4	83	75.2	68
Elongation at break (%)	10.58	7.3	8.7	9.6	8.4	7.1
Impact strength (kJ/m ²)	17	19	21	23	17	14
Hardness, Shore D	78.4	79.2	80.6	81.4	78.8	77.6



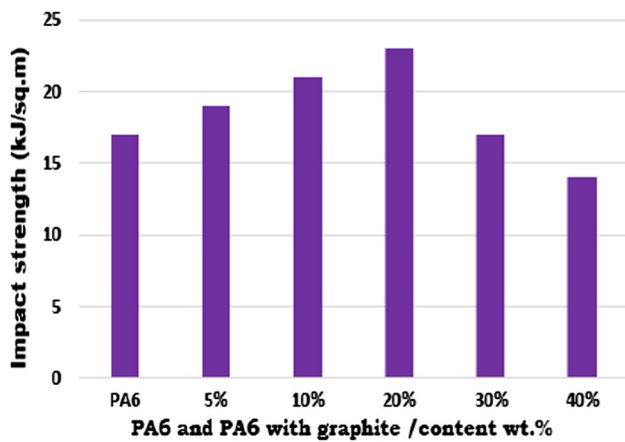


Fig. 9 Impact strength of PA6 and graphite-reinforced PA6 composites

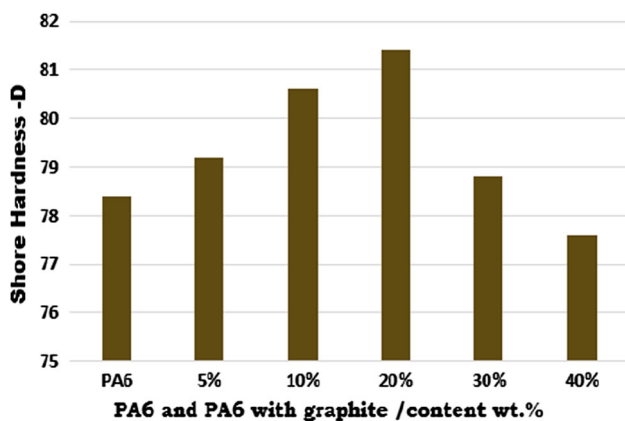


Fig. 10 Hardness of PA6 and graphite-reinforced PA6 composites

tion of graphite in PA6 shows the negative effect on impact strength. From Table 4, graphite-reinforced PA6 composite of 20 wt% has increased the impact strength of 26 % compared with pure PA6.

4.6 Comparison of Hardness Properties for PA6 Composites

Figure 10 shows the results of the hardness values of PA6 and graphite-reinforced PA6 composites. Graphite in contents added to the base polymer (PA6) increases the hardness apparently. The hardness value of graphite-reinforced PA6 composites gradually increased by the addition of graphite content with PA6. The maximum shore hardness value of 81.4 occurred in PA6 + 20 wt% Gr composite. The graphite content increased up to 20 wt%, and the hardness property of PA6 composite has been improved. The improvement in hardness may due to increased surface area of filler in the matrix. On the other hand, as can be suggested from the hardness test, the elastic behavior of the matrix proportionately varies with the addition of PA6 with 30 % graphite parti-

cles. The improved properties of prepared polymer materials and their physical and mechanical properties are reported in Table 4.

5 Conclusion

In this work, PA6 and graphite-reinforced PA6 composites were prepared by using an injection molding machine. The mechanical and tribological properties were observed by conducting tensile, impact, hardness and wear tests. The morphological images of PA6 graphite-reinforced wear test specimen's surfaces under before and after wear test condition were examined in SEM. The incorporation of graphite in PA6 significantly increased the mechanical behavior and wear resistance and decreased the friction coefficient. The uniform distribution and incorporation of the graphite particles in the microstructure of the PA6 material are the major factor responsible for the enhancement of the mechanical and tribological properties [33]. The good affinity between the PA6 and graphite particles is drastically improved the tensile, impact and hardness properties. The mechanical behavior test results show that the addition of graphite particles content had a noticeable effect on reinforcing the PA6 matrix. Moreover, by introducing small amounts of graphite, the mechanical and tribological properties of PA6 composites can be improved tremendously. We found that the composites containing 20 wt% of Gr in the PA6 matrix exhibited the best performance in tensile strength, impact strength, hardness and improved tribological behaviors. Our findings are useful for providing some practical guidance for the preparation of PA6 polymer composites used in engineering fields.

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